Name:_____Period:_____

DISCOVER ESSRT 2

Mineral Composition of Igneous Rocks

Earth & Space Sciences Reference Tables, page 14 · HS-ESS2-3

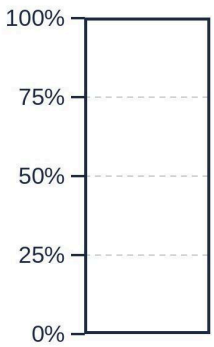
Directions: Follow the instructions to work through each part. Answer the questions in the spaces provided. Parts marked **PAPER** use only your printed Reference Tables. Parts marked **EXPLORER** use the Mineral Composition explorer on the class website.

Vocabulary: felsic, intermediate, mafic, ultramafic, mineral, percent by volume, silica, crystal size, melting temperature

Prior Knowledge Questions

Do these BEFORE you open your Reference Tables or the Explorer.

1. A bag of trail mix is 50% peanuts, 30% raisins, and 20% chocolate. On the bar to the right, draw lines to split it into three parts that show how much of each is in the bag. Label each part.



2. Two rocks are made of exactly the same minerals, in the same amounts, but one has big crystals you can see and the other has crystals too small to see. What could have made them different?

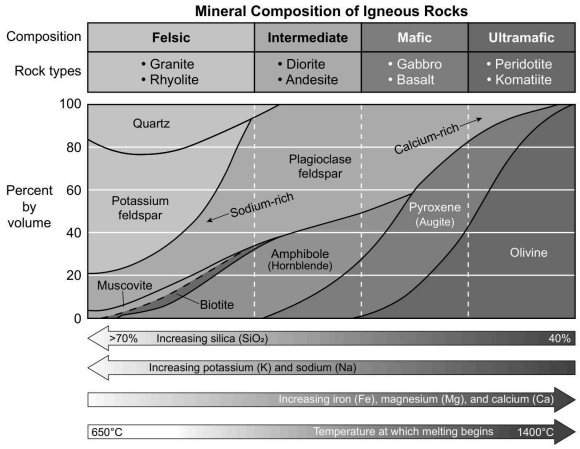
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Warm-up

PAPER · No device needed

Open your printed Reference Tables to **page 14**. The bottom chart (shown below) is **Mineral Composition of Igneous Rocks**. It shows which minerals make up each kind of igneous rock, and how much of each. Before using the Explorer, get to know the chart.



ESSRT page 14, bottom: Mineral Composition of Igneous Rocks (NYSED, 2026 Rev. Edition)

1.Across the top, the chart sorts igneous rocks into four groups. Write each group and its two rock types, left to right.

Group	Rock types

2.Look at the scale on the left side of the graph.

a. What does it measure? It measures:_____

b. What number is at the bottom? Bottom: _____ At the top? Top:_____

3. Each shaded area in the graph is one mineral. How many minerals are named? List them. Number: _____

Minerals:_____

_____

4.Look at the four arrows under the graph.

a. Which direction does the **silica** arrow point, and what values are printed at its ends? Silica points _____ ; ends:_____

b. Which direction does the **melting temperature** arrow point, and what values are printed at its ends? Melting temperature points _____ ; ends:_____

Activity A — How to Read the Chart

Get the Explorer ready:
On the class website, open **Mineral Composition of Igneous Rocks** (page 14) from the Table of Contents, then click **Open the interactive version**.
Click **New to this? Start here**.
Use **Next** and **Back** to move through the steps.

Question:
How do you read how much of each mineral is in a rock?

EXPLORER

1.Read steps 1 and 2. What do the **white dashed lines** in the graph show? _____

2.Read step 3. Granite and rhyolite are both felsic. What do they have in common? How are they different, and why?

In common:_____

Different, because:_____

3.Read step 4.

a. What does each shaded area stand for? _____

b. Which two minerals are thin slivers?_____&_____

4.Read step 5.

a. What does **percent by volume** mean? _____

b. What does the whole height of the graph, 0 to 100, stand for?_____

5.**Record:** Read step 6. Write each mineral the red line crosses and its percent. Then add them up.

Mineral	Percent
Total	

6.Read step 7. Fill in the blanks, then answer the question.

The pyroxene area starts at about _____ and ends at about _____ , so the rock is _____ % pyroxene.

a.Why is the rock **not** 42% pyroxene?_____

7.Read step 8. Toward which side of the chart, **felsic** or **mafic**, does each one increase?

Silica	Potassium and sodium	Iron, magnesium, and calcium

a. Where do the names **felsic** and **mafic** come from?_____

8.Read step 9.

a. At about what temperature do felsic rocks begin to melt? _____

b. Ultramafic rocks? _____

c. Which minerals in this chart are the **first to crystallize** in Bowen's Reaction Series?

First to crystallize: _____ _____

PAPER · Back to the page

9.On your printed page 14, look at the very **left edge** of the graph. Where does the **quartz** area start on the scale? What percent of that rock is quartz? Show your subtraction.

Quartz starts at about _____ : _____ − _____ = _____ %

Activity B — Read Any Rock with the Line

Get the Explorer ready:

Finish the walkthrough (**Try it myself**) or **Close** it.

Drag the **red line** across the chart, or use the **Jump to** buttons above it.

Watch the numbers beside the line and the cards on the side.

Question:

How does what a rock is made of change from felsic to ultramafic?

EXPLORER

1.**Record:** Press each **Jump to** button. Use the **Along the red line** card to fill in the table.

Group	Rocks	Minerals and their percents (from the card)	Total
Felsic			
Intermediate			
Mafic			
Ultramafic			

2.Jump to **Mafic**. In the **Along the red line** card, click **Pyroxene**. The chart traces its bottom and top across to the scale. Record them.

Bottom: _____ Top: _____ Top − bottom = _____ %

3.**Observe:** Drag the line slowly from the left edge of the graph to the right edge.

a. Which minerals disappear along the way? _____

b. Which ones appear?_____

c. Which mineral is in all four groups?_____

4.**Record:** Use the **Along the arrows** card. Write the reading for each arrow (for example, **high** or **lowest**) for a felsic rock and an ultramafic rock.

Arrow	Felsic	Ultramafic
Silica		
Potassium and sodium		
Iron, magnesium, and calcium		
Melting temperature		

5.**Explain:** Mafic rocks are usually darker than felsic rocks. Use your answers to questions 3 and 4 to explain why.

_____

_____

PAPER · Back to the page

6.Use only your printed page 14 and a ruler. Lay the ruler along the **dashed line between the mafic and ultramafic columns**. Estimate the percent of each mineral the line crosses. Show top − bottom for each.

Mineral	Bottom	Top	Percent (top − bottom)
Total			

Activity C — Check Yourself

Get the Explorer ready:	Question:
Click **Check yourself: quiz** near the top of the page. Some questions ask you to **move the red line** and press **Check**. Its numbers stay hidden until you do.	Can you read the chart on your own?

EXPLORER

1. **Try it:** Play one round of the quiz. Score: _____ of 8

2. Choose **two** questions you missed or found hardest. Write the question, the correct answer, and what the chart showed after you answered.

Question (short)	Correct answer	What the chart showed

3. Pick one **Move the red line** question. What did you look for on the chart to decide where to put the line?

4. Play another round (**Try another round**). Did your score change? What did you check more carefully the second time?

Model of Bowen's Reaction Series

Earth & Space Sciences Reference Tables, page 14 · HS-ESS2-3

Directions: Follow the instructions to work through each part. Answer the questions in the spaces provided. Parts marked **PAPER** use only your printed Reference Tables. Parts marked **EXPLORER** use the Bowen's Reaction Series explorer on the class website.

Vocabulary: continuous series, crystal, crystallize, discontinuous series, felsic, intermediate, mafic, magma, mineral, plagioclase feldspar, ultramafic

Prior Knowledge Questions

Do these BEFORE you open your Reference Tables or the Explorer.

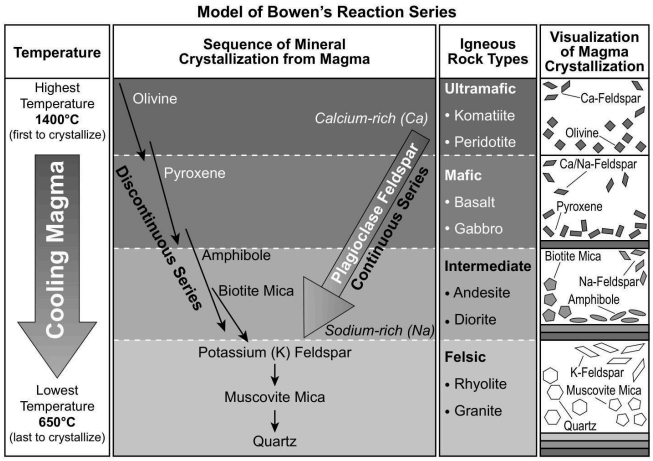
1. Water freezes at 0°C, but melted candle wax turns solid at a much higher temperature. Suppose a hot liquid were a mix of many different substances, each with its own freezing temperature. Predict what would happen as it slowly cooled.

2.Granite is speckled: it has pink, white, gray, and black grains. Each kind of grain is a different mineral. Where do you think those grains came from?

Warm-up

PAPER · No device needed

Open your printed Reference Tables to **page 14**. The top diagram (shown below) is the **Model of Bowen's Reaction Series**. It shows which minerals form, and in what order, as magma (melted rock) cools. Before using the Explorer, get to know the diagram.



ESSRT page 14, top: Model of Bowen's Reaction Series (NYSED, 2026 Rev. Edition)

1.The diagram has four columns. Write the title of each one, left to right.

_____,_____,_____,_____

2.Look at the **Temperature** column.

a. What temperature is printed at the top? At the bottom?

Top: _____ Bottom:_____

b. Which end is labeled "first to crystallize"? _____

3.White dashed lines split the middle of the diagram into rows. How many rows are there? ____

a. Write the rock group named in each row, from top to bottom.

4.Look at the bottom of each box in the **Visualization of Magma Crystallization** column. How many gray stripes are at the bottom of each box, from the top box to the bottom box? What pattern do you notice?

Stripes: _____ , _____ , _____ , _____ Pattern:_____

Activity A — How to Read the Diagram

Get the Explorer ready:	Question:
On the class website, open **Model of Bowen's Reaction Series** (page 14) from the Table of Contents, then click **Open the interactive version**. Click **New to this? Start here**. Use **Next** and **Back** to move through the steps.	How do you read Bowen's Reaction Series, and what do its two branches show?

EXPLORER

1. Read step 1.
   a. What happens to magma's atoms as it cools? _____
   b. What is a **mineral**?_____
2. Read step 2. Which direction do you read the diagram? Why?_____

_____

3. Read step 3 and follow the lit arrows. List the four minerals of the **discontinuous series** in order. Then explain why it is called "discontinuous."

   1st _____ 2nd _____ 3rd _____ 4th

4. Read step 4.
   a. Which mineral makes up the **continuous series**? _____
   b. What changes about it as the magma cools?_____
5. **Compare:** In your own words, explain the difference between the two series.

_____

_____

6. Read step 5. List the last three minerals to crystallize, in order.

   1st _____ 2nd _____ Last

7. **Record:** Read step 6 and read across the lit row. Fill in the table.

Minerals crystallizing	Rock group	Two rocks

8. Read step 7. Write the four rock groups in order from the hottest to the coolest.

(hottest)_____→ _____→ _____→ _____(coolest)

PAPER · Back to the page

9. On your printed page 14, find the words **Calcium-rich (Ca)** and **Sodium-rich (Na)**. Which rock group's row is each one in?

   Calcium-rich (Ca):_____

   Sodium-rich (Na):_____

Activity B — Cool the Magma

Get the Explorer ready:	Question:
Finish or **Close** the walkthrough. Drag the **red line** down the diagram, or press **Cooler** to move one stage at a time. Watch the card on the side: the magma chamber, the rock photos, and the lists under them.	What happens, in order, as one magma cools from 1400°C to 650°C?

EXPLORER

1. **Record:** Move the line through all four stages. For each stage, use the **Crystallizing now** list and the rock photos under **Rocks made mostly of these minerals** to fill in the table.

Stage	Minerals crystallizing now	Rock group	Rocks
1 (hottest)			
2			
3			
4 (coolest)			

2. **Observe:** Watch the magma chamber as you move from stage 1 into stage 2.

   a. What happens to the crystals that formed in stage 1?_____

   b. Move to stage 4. How many gray layers are at the bottom of the chamber?_____

3. **Analyze:** Look at the **What's left in the magma** bars as you cool the magma. Describe how each bar changes. Then use the minerals in your table to explain why.

   Iron, magnesium, calcium:_____

   Silica, potassium, sodium:_____

   Why:_____

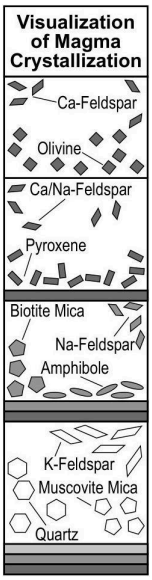
4. Drag the line all the way to the bottom.

   a. What is the last mineral to crystallize?_____

   b. What is left of the magma?_____

PAPER · Back to the page

Detail of page 14: the Visualization column



5. Use only your printed page 14. A rock is made mostly of **amphibole**, **biotite mica**, and **sodium-rich plagioclase feldspar**.

   a. Which rock group does it belong to? Name two rocks in that group.

      Rock group: _____ Rocks:_____&_____

   b. Did its minerals crystallize at a higher or lower temperature than the minerals in **gabbro**?

   _____ temperature

   c. How can you tell from the diagram?_____

   _____

Activity C — Check Yourself

Get the Explorer ready:	Question:
Click **Check yourself: quiz** near the top of the page. Answer each question using the diagram. After you answer, the diagram lights up where to find it.	Can you use the diagram to answer questions on your own?

EXPLORER

1. **Try it:** Play one round of the quiz. Score: _____ of 8

2. Choose **two** questions you missed or found hardest. Write the question, the correct answer, and what part of the diagram lit up to show it.

Question (short)	Correct answer	Where it is on the diagram

3. Play another round (**Try another round**). Did your score change? What did you check more carefully the second time?

Build It Better

Your Feedback on the Explorer

The Explorer is a **model** of your printed Reference Tables: the same data, shown a new way. Scientists and engineers test their models, find the strengths and limitations, and improve them. That's your job on this page. Your teacher is working with a company that builds the Explorer by coding, and feedback like yours is what tells it what to change in the next version.

No name needed.

Part of the Explorer: Mineral Composition of Igneous Rocks: walkthrough, line reader, and quiz (page 14)

Device: □ Chromebook □ Laptop □ Phone □ Tablet

1. **Rate it.** Circle one number for each.

	1 No	2	3	4 Yes!
It was easy to figure out what to click.	1	2	3	4
It helped me understand the printed page.	1	2	3	4
It worked well on the device I used.	1	2	3	4
I would use it again to study.	1	2	3	4

2. **Two models, one set of data.** Compare the Explorer with the printed page.

+ **Strength.** Something the Explorer shows **better** than the printed page:	– **Limitation.** Something the printed page does better, or something the Explorer **gets wrong or leaves out**:

3. **Fix one thing.** Be specific enough that someone could build it from what you write.

What kind?	□ Something broke □ Something was confusing □ An idea for something new
Where? Which step, screen, or button	
What happened? Or: what's missing	
What should happen instead?	

4. **Design it.** Sketch your change as it would look on the screen, and label it.

For teachers: NYSSLS Science & Engineering Practices: Developing and Using Models (evaluate the merits and limitations of a model) and Engineering Design (evaluate a solution against criteria; refine it). HS-ETS1-3.